

Toghrul Karimov

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I am a theoretical computer scientist working at the intersection of algorithmic problems of dynamical systems, automata theory, logic, and number theory.

Academic employment¹

04/2025 - present	Postdoctoral researcher working with Valérie Berthé (Institut de Recherche en Informatique Fondamentale, Paris, France) Funded by the ERC Synergy Grant “DnyAMiCs”
03/2024 - 03/2025	Postdoctoral researcher working with Joël Ouaknine (Max Planck Institute for Software Systems, Saarbrücken, Germany)

Education

2019-2024	PhD student at Saarland University and the Max Planck Institute for Software Systems, Saarbrücken, Germany Supervisor: Joël Ouaknine Thesis: Algorithmic verification of linear dynamical systems doi Awarded the Ackermann Award 2025 and the Otto Hahn Medal of the Max Planck Society
2015-2019	MCompSci Computer Science, University of Oxford, UK First Class Honours

Scholarships and awards

8. **Otto Hahn Medal** of the Max Planck Society for my PhD thesis and contributions as a junior scientist, 2026.
7. **EACSL Outstanding Dissertation Award** for Logic in Computer Science (Ackermann Award) 2025.
6. **Fellowship** of the Hausdorff Research Institute for Mathematics (HIM) in Bonn to attend the Trimester Program “Definability, Decidability, and Computability”, October 2025.
5. **ACM-SIGBED Best Paper Award** at the conference Hybrid Systems: Computation and Control (HSCC) 2024 for the paper “Linear dynamical systems with continuous weight functions” together with R. Aghamov, C. Baier, J. Ouaknine, and J. Piribauer.

¹I am currently formally employed by the Max Planck Institute for Software Systems in Saarbrücken, Germany, and paid by an ERC Synergy grant between Paris and Saarbrücken. I travel regularly to Paris, where I work with V. Berthé and others.

4. **Distinguished Paper Award** at the conference ACM/IEEE Symposium on Logic in Computer Science (LICS) 2024 for the paper “On the decidability of monadic second-order logic with arithmetic predicates” together with V. Berthé, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell.
3. CPEC (see www.perspicuous-computing.science) **mini-project award** for a two-week research visit to University of Oxford, June 2022.
2. Keble College **Scholarship**, 2016-2019. Awarded for excellent academic performance at the end of each year.
1. The **Scholarship** of the Ministry of Education of Azerbaijan covering the full costs of my study at the University of Oxford, 2015-2019.

Professional service

- 35th EACSL Annual Conference on Computer Science Logic 2027, member of the program committee
- Regular reviewer for the conferences LICS, ICALP, MFCS, STOC, FOCS, SODA

Teaching and supervision²

05/2026-07/2026	Supervising Daria Goliasova, a research intern at the MPI-SWS. We are building an efficient implementation of this result using an SMT solver. Here is a preliminary version.
2026	“Principles of cybersecurity”, lecturer, also responsible for practicals and tutorials. Azerbaijan University of Architecture and Construction (online) lecture notes
2026	“Machine learning”, lecturer (also responsible for practicals and tutorials) Azerbaijan University of Architecture and Construction (online) lecture notes
2022	“Topics in algorithmic dynamical systems theory”, teaching assistant Saarland University
2020	“Automata and sequences”, teaching assistant Saarland University

Invited talks and presentations

19. Automata on S -adic words (seminar talk). *One World Numeration Seminar*, online, 2026. [seminar website](#)
18. Rich sequences and decidability of logical theories (conference talk). *Mathematical and Algorithmic Aspects of Dynamical Systems*, Bellairs Research Institute, Barbados, 2026. [conference website](#)
17. Algorithmic verification of linear dynamical systems (thesis presentation). *Computer Science Logic (CSL)*, Paris, France, 2026. [slides](#)

²Teaching AI and cybersecurity was not part of my official job as a researcher: I established two important courses from zero at an Azerbaijani university to gain teaching experience and also as a “human development project”.

16. Decidability of logical theories via rigidity and randomness in dynamical systems (conference talk). *Algebraic Methods in Dynamics and Particle Physics*, Saarbrücken, Germany, 2025. [abstract](#) and [slides](#)
15. Some applications of o-minimality to computational problems in dynamical systems theory (seminar talk). *Hausdorff Research Institute for Mathematics*, Bonn, Germany, 2025. [video recording](#)
14. Arithmetic predicates and decidability of logical theories (invited talk). *Hausdorff Research Institute for Mathematics*, Bonn, Germany, 2025. [video recording](#)
13. Extensions of linear programming with squares and powers of integers (invited talk). *Kyoto University*, Japan, 2025.
12. Rich sequences and decidability of logical theories (conference talk). *Numeration and Substitution 2025*, Tsukuba, Japan. [conference website](#)
11. Ergodicity for linear dynamical systems via o-minimality (conference talk). *Highlights of Logic, Games, and Automata 2025*, Saarbrücken, Germany. [conference website](#)
10. From word combinatorics to automatic structures (invited talk), *Workshop on Recent Developments in Arithmetic Theories and Applications*, Kolkata, India, 2025. [conference website](#)
9. On the decidability of Presburger arithmetic expanded with powers (paper presentation). *Symposium on Discrete Algorithm (SODA) 2025*, New Orleans, United States. [abstract](#)
8. Ode to o-minimality (invited talk). *Symbolic Dynamics and Arithmetic Expansions*, Roscoff, France, and *Stellenbosch University Logic Seminar*, South Africa, 2024. [slides](#)
7. The power of Positivity (paper presentation). *Logic in Computer Science (LICS) 2023*, Boston, United States. [conference website](#)
6. The model-checking problem for linear dynamical systems (conference talk). *Workshop on Algorithmic Aspects of Dynamical Systems*, Barbados, 2023. [conference website](#)
5. The pseudo-reachability problem for diagonalisable affine dynamical systems (paper presentation). *Mathematical Foundations of Computer Science (MFCS) 2022*, Vienna, Austria, and *Reachability Problems (RP) 2022*, Saarbrücken, Germany. [conference website](#), [conference website](#)
4. The pseudo-Skolem problem is decidable (paper presentation). *Mathematical Foundations of Computer Science (MFCS) 2021*, Tallinn, Estonia. [conference website](#)
3. Invariants and impossibility: from geometric constructions to solving polynomial equations. *Monsoon Math 2021*, an online camp for Indian students.
2. Deciding ω -regular properties on linear recurrence sequences (paper presentation). *Principles of Programming Languages (POPL) 2021*, online. [abstract](#)
1. On LTL model-checking for low-dimensional discrete linear dynamical systems (paper presentation). *Mathematical Foundations of Computer Science (MFCS) 2020*, online. [conference website](#)

Publications in reverse chronological order³

21. *Automata on S -adic words [arXiv](#)

V. Berthé, T. Karimov, and M. Vahanwala

International Colloquium on Automata, Languages, and Programming (ICALP) 2026

³In theoretical computer science, the authors of a paper are usually listed in the alphabetical order. The papers in which Toghrul Karimov was the clear lead author are marked with an asterisk.

20. *Linear dynamical systems with weight functions [doi](#)
R. Aghamov, C. Baier, T. Karimov, J. Piribauer, and J. Ouaknine
Nonlinear Analysis: Hybrid Systems, Vol. 60
19. Model checking linear temporal logic with standpoint modalities [doi](#)
R. Aghamov, C. Baier, T. Karimov, R. Majumdar, J. Ouaknine, J. Piribauer, and T. Spork
Principles of Knowledge Representation and Reasoning (KR) 2025
18. Multiple reachability in linear dynamical systems [doi](#)
T. Karimov, E. Kelmendi, J. Ouaknine, and J. Worrell
Logic in Computer Science (LICS) 2025
17. Algorithmic applications of Schanuel’s conjecture [doi](#)
T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala and J. Worrell
Principles of Formal Quantitative Analysis: Essays Dedicated to Christel Baier on the Occasion of Her 60th Birthday
16. *Verification of linear dynamical systems via o-minimality of the real numbers [doi](#)
T. Karimov
International Colloquium on Automata, Languages, and Programming (ICALP) 2025
15. On the decidability of Presburger arithmetic expanded with powers [doi](#)
T. Karimov, F. Luca, J. Nieuwveld, J. Ouaknine, and J. Worrell
Symposium on Discrete Algorithms (SODA) 2025
14. *The monadic theory of toric words [doi](#)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell
Theoretical Computer Science, Vol. 1025
13. On the decidability of monadic second-order logic with arithmetic predicates [doi](#)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell
Logic in Computer Science (LICS) 2024
Distinguished Paper Award
12. *Linear dynamical systems with continuous weight functions [doi](#)
R. Aghamov, C. Baier, T. Karimov, J. Piribauer, and J. Ouaknine
Hybrid Systems: Computation and Control (HSCC) 2024
ACM SIGBED Best Paper Award

11. Model checking Markov chains as distribution transformers [doi](#)
R. Aghamov, C. Baier, T. Karimov, J. Nieuwveld, J. Ouaknine and M. Vahanwala
Principles of Verification: Cycling the Probabilistic Landscape. Essays Dedicated to Joost-Pieter Katoen on the Occasion of His 60th Birthday, Part II
10. *The power of Positivity [doi](#)
T. Karimov, E. Kelmendi, J. Nieuwveld, J. Ouaknine and J. Worrell
ACM/IEEE Symposium on Logic in Computer Science (LICS) 2023
9. *What’s decidable about discrete linear dynamical systems? [doi](#)
T. Karimov, E. Kelmendi, J. Ouaknine and J. Worrell
Principles of Systems Design: Essays Dedicated to Thomas A. Henzinger on the Occasion of His 60th Birthday
8. Parameter synthesis for parametric probabilistic dynamical systems and prefix-independent specifications [doi](#)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefauchaux, J. Ouaknine, D. Purser, M. Whiteland, and J. Worrell
International Conference on Concurrency Theory (CONCUR) 2022
7. *The pseudo-reachability problem for diagonalisable linear dynamical systems [doi](#)
J. D’Costa, T. Karimov, R. Majumdar, J. Ouaknine, M. Salamati, and J. Worrell
Mathematical Foundations of Computer Science (MFCS) 2022
6. *What’s decidable about linear loops? [doi](#)
T. Karimov, E. Lefauchaux, J. Ouaknine, D. Purser, J. Worrell, and M. Whiteland
POPL 2022, Proceedings of the ACM on Programming Languages, Vol. 6
5. The orbit problem for parametric linear dynamical systems [doi](#)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefauchaux, F. Luca, J. Ouaknine, D. Purser, M. Whiteland, and J. Worrell
International Conference on Concurrency Theory (CONCUR) 2021
4. *The pseudo-Skolem problem is decidable [doi](#)
J. D’Costa, T. Karimov, R. Majumdar, J. Ouaknine, M. Salamati, S. Soudjani, and J. Worrell
Mathematical Foundations of Computer Science (MFCS) 2021
3. *Deciding ω -regular properties on linear recurrence sequences [doi](#)
S. Almagor, T. Karimov, E. Kelmendi, J. Ouaknine, and J. Worrell
POPL 2021, Proceedings of the ACM on Programming Languages, Vol. 5

2. Reachability in dynamical systems with rounding [doi](#)
C. Baier, F. Funke, S. Jantsch, T. Karimov, E. Lefauchaux, J. Ouaknine, A. Pouly, D. Purser, and M. Whiteland
Foundations of Software Technology and Theoretical Computer Science (FSTTCS) 2020
1. *On LTL model-checking for low-dimensional discrete linear dynamical systems [doi](#)
T. Karimov, J. Ouaknine, and J. Worrell
Mathematical Foundations of Computer Science (MFCS) 2020

Papers under peer review

- A trichotomy for monadic theories of linear recurrence sequences [pdf](#)
T. Karimov, J. Nieuwveld, J. Ouaknine
- Preservation theorems for transducers outputs [pdf](#)
V. Berthé, H. Goulet-Oullet, T. Karimov, D. Perrin, and M. Vahanwala
- *Rich sequences and decidability of logical theories [pdf](#)
T. Karimov, J. Nieuwveld, and J. Ouaknine
- On the decidability of monadic theories of arithmetic predicates [arXiv](#)
V. Berthé, T. Karimov, J. Nieuwveld, J. Ouaknine, M. Vahanwala, and J. Worrell